TB. The pathogenesis of Primary Tuberculosis is described in flowchart 7.8.

Source of infection – Airborne droplets

Route of entry – Respiratory tract

Incubation period – 3–6 weeks.

Tubercle bacilli enter the host commonly by inhalation. When bacilli are inhaled, the bacilli enters into the alveolar macrophage, where they can grow and multiply.

Non-resident macrophage are also attracted to the site of infection and these macrophages are engulfs the fubercle bacilli. The bacilli were carried through the lymphatic to the local hilar lymph node.

In the lymph nodes, cell mediated immunity (CMI) is stimulated. The CMI response helps to prevent the further spread of pathogen.

Within 10 days of infection, T-lymphocytes produces lymphokines which activate macrophages and leads to form granuloma, around the foci of infection.

The activated marophages are termed epithelioid cells, which fuse to form multinucleate giant cells.

The granuloma contains necrotic tissue and dead macrophage which is referred as caseation. Granuloma formation is usually sufficient to limit the primary infection.

The lung lesions is frequently found in the lower lobe and called as ghon focus. The ghon focus together with enlarged hilar lymph nodes is called primary complex.

In some persons, the lesions become dormant and produce dense scar tissue which may become calcified. Some bacilli remain in a dormant form as persisters which, when reactivated, cause post primary TB.

In a minority of cases, the ghon focus ruptures into a blood vessel. Then the bacilli spread throughout the body with the formation of numerous granulomas and known as miliary TB.

Flowchart 7.8: The pathogenesis of Primary Tuberculosis

Secondary TB – (Post primary TB) It is caused by reactivation of the primary lesion or by exogenous reinfection. Granulomas of secondary TB most often occur in the apex of the lungs. The necrotic element of the reaction causes tissue destruction and the formation of large area of caseation termed tuberculomas. The presence of caseous necrosis and cavities are two important clinical manifestations of secondary TB. The cavities may rupture into blood vessels, spreading the bacilli throughout the body and break into airways, releasing the pathogen in aerosols and sputum - called as open tuberculosis (Figure 7.20).

7.11.4 Clinical Symptoms

It includes, cough that lasts for more than 2–3 weeks, weight loss, fever, night sweat and loss of appetite.

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